

Programming in Python for Friends and Relations—Part 15

Personalising Photographs



Most of us enjoy sharing our photographs. We'd also like to add captions that can't be missed or send postcard-size images. That's what the Python imaging library is all about!

To manipulate individual images, products like The GIMP are most appropriate. However, if similar actions need to be applied to a group of images, consider programming them using the Python imaging module.

In order to keep the code simple, the assumption is that you have just downloaded the pictures from your camera. So, all the pictures you wish to process are in the same directory. Create a sub-directory *save/* in which you will keep your processed photos.

After that, we will go over how to write a Python generator and use the imaging library to create image transition effects when viewing photographs.

Transformation of images

Write a class called *photos* to track all the files in the current directory. You will get one photo at a time and can resize it to a pre-defined size. You can then modify this

photo and add a caption to it. Finally, you can save the modified photograph in the *save/* directory.

```
import os
import Image, ImageTk, ImageDraw
class photos:
    def __init__(self, new_size):
        self.file_generator = (fn for fn in os.listdir('.'))
        self.new_size = new_size
        self.image = None
```

The initialisation method is simple enough. The *file_generator* will be convenient for fetching the next file when needed. You can add the flexibility of passing the directory as a parameter.

```
def get_next_photo(self):
    while True:
        try:
            self.file_name = self.file_generator.next()
            image = Image.open(self.file_name)
```